

MATH70078 Fundamentals of Statistical Inference (2025/26)

Coursework Solutions

Deadline: November 7, 12:00 PM (noon) on Turnitin

MSc Statistics, Imperial College London

CID: 06060027

Submitted: 7 November 2025

Problem 1

(1.1) Bayes action without data; Bayes risk with data

Solution. When no data are available, the decision maker must fix a single action $a \in A$. The Bayes action a_0 minimises the prior expected loss:

$$R_0(\pi) = \min_{a \in A} \int_{\Theta} L(\theta, a) \pi(\theta) d\theta, \quad a_0 \in \arg \min_a \int_{\Theta} L(\theta, a) \pi(\theta) d\theta.$$

The associated minimal Bayes risk is

$$R_0 = \int_{\Theta} L(\theta, a_0) \pi(\theta) d\theta.$$

Once data Y are observed, a decision rule $d(\cdot)$ assigns an action $d(y)$ to each realisation of Y , and

$$R_1 = \iint L(\theta, d(y)) \pi(\theta) f(y; \theta) dy d\theta,$$

where $f(y; \theta)$ is the conditional density of Y given θ .

Using the joint density $p(\theta, y) = \pi(\theta)f(y; \theta)$ and Fubini's theorem, this can be written as

$$R_1 = \int_Y f(y) \left[\int_{\Theta} L(\theta, d(y)) \pi(\theta | y) d\theta \right] dy,$$

where the posterior $\pi(\theta | y) = p(\theta, y)/f(y)$ and $f(y) = \int_{\Theta} p(\theta, y) d\theta$.

For each fixed y , the optimal Bayes rule $d^*(y)$ minimises the posterior expected loss,

$$r(a; y) = \int_{\Theta} L(\theta, a) \pi(\theta | y) d\theta, \quad d^*(y) \in \arg \min_a r(a; y),$$

so that

$$R_1 = \int_Y f(y) \left[\min_a \int_{\Theta} L(\theta, a) \pi(\theta | y) d\theta \right] dy.$$

Comparison of R_1 and R_0 . Consider the two quantities:

$$(i) \quad \min_a \int_{\Theta} L(\theta, a) \pi(\theta | y) d\theta \quad \text{and} \quad (ii) \quad \int_{\Theta} L(\theta, a_0) \pi(\theta | y) d\theta,$$

where a_0 is the prior-optimal action. Since (i) is the minimum of the posterior expected loss over all a , it is necessarily no larger than (ii):

$$\min_a \int_{\Theta} L(\theta, a) \pi(\theta | y) d\theta \leq \int_{\Theta} L(\theta, a_0) \pi(\theta | y) d\theta \quad \text{for all } y.$$

Multiplying by the nonnegative weight $f(y)$ and integrating over y gives

$$\int_Y f(y) \left[\min_a \int_{\Theta} L(\theta, a) \pi(\theta | y) d\theta \right] dy \leq \int_Y f(y) \int_{\Theta} L(\theta, a_0) \pi(\theta | y) d\theta dy.$$

The right-hand side can be simplified by applying Fubini and the law of total probability:

$$\begin{aligned} \int_Y f(y) \int_{\Theta} L(\theta, a_0) \pi(\theta | y) d\theta dy &= \int_Y \int_{\Theta} L(\theta, a_0) \frac{\pi(\theta) f(y | \theta)}{f(y)} f(y) d\theta dy \\ &= \int_Y \int_{\Theta} L(\theta, a_0) \pi(\theta) f(y | \theta) d\theta dy \\ &\stackrel{\text{Fubini}}{=} \int_{\Theta} L(\theta, a_0) \pi(\theta) \left[\int_Y f(y | \theta) dy \right] d\theta \\ &= \int_{\Theta} L(\theta, a_0) \pi(\theta) d\theta = R_0. \end{aligned}$$

Since $\int_Y f(y | \theta) dy = 1$ for each θ . Hence,

$$\boxed{R_1 = \int_Y f(y) \left[\min_a \int_{\Theta} L(\theta, a) \pi(\theta | y) d\theta \right] dy \leq R_0.}$$

Explanation of the inequality. This result can also be viewed as an application of the convexity property of expectation: for any family of random variables $\{Z_a\}$,

$$\mathbb{E}[\min_a Z_a] \leq \min_a \mathbb{E}[Z_a].$$

Here $Z_a = L(\Theta, a)$ and the outer expectation is over the joint law of (Θ, Y) . Intuitively, the left-hand side corresponds to choosing the best action *after* observing Y , while the right-hand side fixes one action before seeing any data. Because the minimum operator is convex, minimising inside the expectation can only decrease the overall expected loss.

Interpretation. Conditioning on Y refines our knowledge of Θ , so posterior-based decisions adapt to the realised data and cannot perform worse on average than any fixed rule. The strict inequality $R_1 < R_0$ holds whenever the data are informative (that is, $\pi(\theta | y) \neq \pi(\theta)$ for a set of y with positive probability).

Quantifying the gain from information. Finally, the reduction in Bayes risk can be expressed as

$$R_0 - R_1 = \int_Y f(y) \left[\int_{\Theta} L(\theta, a_0) \pi(\theta | y) d\theta - \min_a \int_{\Theta} L(\theta, a) \pi(\theta | y) d\theta \right] dy.$$

The integrand measures, for each y , the improvement obtained by switching from the fixed prior-optimal action a_0 to the posterior-optimal one. Averaging this improvement with respect to $f(y)$ yields the overall *value of information*: the expected reduction in Bayes risk due to conditioning on the data.

(1.2) Highest-density set under loss $L(\theta, S) = c|S| - \mathbf{1}_S(\theta)$

Fix $0 < c < 1/\sqrt{2\pi}$ and let the action be a Borel set $S \subset \mathbb{R}$ with loss $L(\theta, S) = c|S| - \mathbf{1}_S(\theta)$, where $|S|$ denotes the Lebesgue measure. Without data (so that decisions are made under the prior π), the Bayes criterion is

$$\int L(\theta, S) \pi(\theta) d\theta = c|S| \int_{\Theta} \pi(\theta) d\theta - \int_{\Theta} \mathbf{1}_S(\theta) \pi(\theta) d\theta = c|S| - \int_S \pi(\theta) d\theta.$$

Minimizing this quantity over all measurable S is equivalent to *maximizing*

$$\int_S \pi(\theta) d\theta - c|S| = \int_S (\pi(\theta) - c) d\theta.$$

To identify the maximizing set, note that the integrand in $\int_S (\pi(\theta) - c) d\theta$ is positive when $\pi(\theta) > c$ and negative when $\pi(\theta) < c$. If S contains any region where $\pi(\theta) < c$, removing that region would strictly increase the value of the integral, whereas enlarging S within $\{\pi(\theta) > c\}$ would increase it further. Hence, the maximiser must include all points where $\pi(\theta) > c$ and exclude all where $\pi(\theta) < c$ (points with $\pi(\theta) = c$ being indifferent). Therefore the optimal set (unique up to null sets) is the *highest-density region*

$$S^* = \{\theta : \pi(\theta) \geq c\}.$$

Hence S^* is the Bayes action: it minimizes the prior expected loss $\int L(\theta, S) \pi(\theta) d\theta$.

The corresponding Bayes risk is

$$R_0(\pi) = c|S^*| - \int_{S^*} \pi(\theta) d\theta = \int_{S^*} (c - \pi(\theta)) d\theta \leq 0,$$

because $\pi(\theta) \geq c$ on S^* (and $R_0 = 0$ only if $\pi(\theta) \equiv c$ on S^*). We have thus shown that the *highest-density region* S^* is the Bayes action for this decision problem and that the associated Bayes risk is non-positive.

(1.3) Prior $\mathcal{N}(\mu, \sigma^2)$: explicit expression of R_0 in terms of Φ

Let the prior density be

$$\pi(\theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(\theta - \mu)^2}{2\sigma^2}\right\}.$$

The optimal set is $S^* = \{\theta : \pi(\theta) \geq c\}$. We now solve explicitly the inequality

$$\frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(\theta - \mu)^2}{2\sigma^2}\right\} \geq c.$$

Since $c > 0$, taking natural logarithms on both sides yields

$$-\frac{(\theta - \mu)^2}{2\sigma^2} \geq \log(c\sqrt{2\pi}\sigma) \iff (\theta - \mu)^2 \leq 2\sigma^2 \log\left(\frac{1}{c\sqrt{2\pi}\sigma}\right).$$

Condition for a nonempty level set. The normal density achieves its maximum at $\theta = \mu$, with

$$\pi_{\max} = \pi(\mu) = \frac{1}{\sqrt{2\pi}\sigma}.$$

Hence, the inequality $\pi(\theta) \geq c$ admits real solutions only if

$$0 < c \leq \frac{1}{\sqrt{2\pi}\sigma}.$$

If $c = \pi_{\max}$, the level set reduces to the single point $\{\mu\}$. If $c > \pi_{\max}$, the inequality has no solution and $S^* = \emptyset$. Therefore, for the level set to be nonempty and of positive measure, we must have

$$0 < c < \frac{1}{\sqrt{2\pi}\sigma}.$$

Characterisation of S^* . Define

$$z_c := \sqrt{2 \log\left(\frac{1}{c\sqrt{2\pi}\sigma}\right)} \quad (\text{nonnegative, defined for } 0 < c \leq \frac{1}{\sqrt{2\pi}\sigma}).$$

so that

$$S^* = [\mu - \sigma z_c, \mu + \sigma z_c], \quad |S^*| = 2\sigma z_c.$$

Integral of the prior over S^* . Using the substitution $t = (\theta - \mu)/\sigma$ (so that $d\theta = \sigma dt$), and denoting by φ and Φ the standard normal pdf and cdf respectively, we have

$$\int_{S^*} \pi(\theta) d\theta = \int_{\mu - \sigma z_c}^{\mu + \sigma z_c} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(\theta - \mu)^2}{2\sigma^2}\right\} d\theta = \int_{-z_c}^{z_c} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt = \Phi(z_c) - \Phi(-z_c) = 2\Phi(z_c) - 1.$$

Bayes risk. From the expression derived in (1.2),

$$R_0(\pi) = c|S^*| - \int_{S^*} \pi(\theta) d\theta,$$

so that

$$R_0(\pi) = 2c\sigma z_c - (2\Phi(z_c) - 1), \quad z_c = \sqrt{2 \log\left(\frac{1}{c\sqrt{2\pi}\sigma}\right)}, \quad 0 < c < \frac{1}{\sqrt{2\pi}\sigma}.$$

The parameter z_c represents the half-width, in standard deviation units, of the interval where the prior density exceeds the threshold c . As c decreases, z_c grows and the region S^* widens. The corresponding Bayes risk $R_0(\pi)$ combines the “size penalty” $2c\sigma z_c$ and the probability mass $2\Phi(z_c) - 1$ contained within the interval.

(1.4) Data $Y \mid \theta \sim \mathbf{N}(\theta, 1/9)$: show that $R_1 < R_0$

Posterior distribution. By conjugacy, the posterior of θ given $Y = y$ is Gaussian:

$$\theta \mid Y = y \sim \mathbf{N}(m, v), \quad v = \left(\sigma^{-2} + 9\right)^{-1}, \quad m = v(\sigma^{-2}\mu + 9y).$$

In the given case $\mu = 0$ and $\sigma^2 = 1$, we have

$$v = \frac{1}{10}, \quad m = 0.9y.$$

Hence the posterior is more concentrated around m than the prior is around μ , because $v < \sigma^2$.

Posterior Bayes action. For the same threshold-loss setup as before, the posterior Bayes rule is the highest-posterior-density (HPD) set:

$$S_y^* = \{\theta : \pi(\theta \mid y) \geq c\} = [m - \sqrt{v} z_{c,v}, m + \sqrt{v} z_{c,v}], \quad z_{c,v} := \sqrt{2 \log\left(\frac{1}{\sqrt{2\pi v} c}\right)}.$$

The structure of S_y^* mirrors that of the prior HPD region $S^* = [\mu - \sigma z_{c,\sigma}, \mu + \sigma z_{c,\sigma}]$ from (1.3), but scaled by $\sqrt{v}$ and centred at m instead of μ .

Posterior risk conditional on y . The conditional Bayes loss for this rule is

$$c|S_y^*| - \Pr(\theta \in S_y^* \mid y) = 2c\sqrt{v} z_{c,v} - (2\Phi(z_{c,v}) - 1).$$

Note that this expression depends on v and c but not on y : the location parameter m cancels out due to the symmetry of the normal density. Therefore the posterior (integrated) risk is

$$R_1 = 2c\sqrt{v} z_{c,v} - (2\Phi(z_{c,v}) - 1), \quad v = \frac{1}{10}.$$

Comparison with the prior risk. The prior risk from (1.3) is

$$R_0 = 2c\sigma z_{c,\sigma} - (2\Phi(z_{c,\sigma}) - 1), \quad z_{c,\sigma} = \sqrt{2 \log\left(\frac{1}{c\sqrt{2\pi}\sigma}\right)}.$$

To compare R_1 and R_0 , observe that both risks depend on a single “scale” parameter, say τ (either σ or $\sqrt{v}$), through two competing effects:

- a *linear term* $2c\tau z_{c,\tau}$ that grows with τ , penalising longer intervals;
- a *coverage term* $2\Phi(z_{c,\tau}) - 1$ that increases with $z_{c,\tau}$ and represents the probability mass captured by the HPD region.

Both $z_{c,\sigma}$ and $z_{c,v}$ satisfy the defining relation

$$c = \frac{1}{\sqrt{2\pi}\tau} e^{-z_{c,\tau}^2/2}, \quad \text{that is,} \quad c\sqrt{2\pi}\tau e^{z_{c,\tau}^2/2} = 1.$$

Since $\sqrt{v} < \sigma$ and $\varphi(z) = (2\pi)^{-1/2} e^{-z^2/2}$ is decreasing for $z > 0$, this identity implies

$$\sqrt{v} \varphi(z_{c,v}) = \sigma \varphi(z_{c,\sigma}) = c, \quad \Rightarrow \quad z_{c,v} > z_{c,\sigma}.$$

Intuitively, the narrower posterior density (smaller variance v) must reach a larger standardized threshold $z_{c,v}$ to fall to the same density level c .

Differentiating the risk function

$$R(\tau) = 2c\tau z_{c,\tau} - (2\Phi(z_{c,\tau}) - 1), \quad z_{c,\tau} = \sqrt{2 \log\left(\frac{1}{c\sqrt{2\pi\tau}}\right)},$$

with respect to the scale parameter (for fixed c) shows that $R(\tau)$ is strictly increasing in τ . Hence a smaller scale (a more concentrated distribution) leads to a smaller risk:

$$\sqrt{v} < \sigma \implies R_1 = R(\sqrt{v}) < R(\sigma) = R_0.$$

The inequality $R_1 < R_0$ means that incorporating the data reduces the Bayes risk. The posterior distribution is more concentrated around its mean, so its HPD region of density threshold c captures probability mass more efficiently (that is, more probability per unit length) than the corresponding prior region. The improvement is strict whenever the data are informative (i.e. the posterior differs nontrivially from the prior).

This result illustrates the general decomposition of the Bayes risk:

$$R_0 - R_1 = \mathbb{E}[\text{prior risk} - \text{posterior risk}] = \mathbb{E}[\text{expected reduction from data}] \geq 0,$$

with equality only if the decision rule ignores the data. Therefore, by both explicit calculation and the general Bayes–decision principle,

$$\boxed{R_1 < R_0.}$$

Problem 2. Hard-Threshold Estimation

Let $Y_1, \dots, Y_n$ be independent random variables with

$$Y_i \sim \mathcal{N}(\theta_i, 1), \quad i = 1, \dots, n,$$

and consider estimation of $\theta = (\theta_1, \dots, \theta_n)$ under squared error loss

$$L(\theta, \hat{\theta}) = \sum_{i=1}^n (\hat{\theta}_i - \theta_i)^2.$$

The *hard-threshold* estimator is defined componentwise as

$$\hat{\theta}_{H,i} = \begin{cases} Y_i, & |Y_i| > \lambda, \\ 0, & |Y_i| \leq \lambda, \end{cases} \quad \lambda > 0.$$

(2.1) Tail bound: $P(|X| > t) \leq \frac{2\varphi(t)}{t}$ for $X \sim \mathcal{N}(0, 1)$

Let $X \sim \mathcal{N}(0, 1)$ with pdf $\varphi(x) = (2\pi)^{-1/2}e^{-x^2/2}$ and cdf $\Phi(x)$. For $t > 0$,

$$1 - \Phi(t) = \int_t^\infty \varphi(x) dx.$$

Since $\varphi'(x) = -x\varphi(x)$, for $x > 0$ we can write $\varphi(x) = -\frac{1}{x}\varphi'(x)$. We now apply *integration by parts* to the integral, using the rule

$$\int u dv = [uv] - \int v du.$$

Set

$$u = -\frac{1}{x}, \quad dv = \varphi'(x) dx, \quad du = \frac{1}{x^2} dx, \quad v = \varphi(x).$$

Then

$$\int_t^\infty \left(-\frac{1}{x}\right) \varphi'(x) dx = \left[-\frac{\varphi(x)}{x}\right]_t^\infty - \int_t^\infty \frac{\varphi(x)}{x^2} dx.$$

Hence

$$1 - \Phi(t) = \frac{\varphi(t)}{t} - \int_t^\infty \frac{\varphi(x)}{x^2} dx \leq \frac{\varphi(t)}{t},$$

since $\int_t^\infty \frac{\varphi(x)}{x^2} dx$ is positive.

Then, by symmetry of the normal distribution,

$$P(|X| > t) = 2\{1 - \Phi(t)\} \leq \frac{2\varphi(t)}{t}, \quad t > 0.$$

Thus we proved the desired tail bound:

$$P(|X| > t) \leq \frac{2\varphi(t)}{t} \quad \text{for } X \sim \mathcal{N}(0, 1) \text{ and } t > 0.$$

(2.2) Consistency of the hard-threshold rule under $\theta \equiv 0$

Assume $\theta_i \equiv 0$ for all $i = 1, \dots, n$, so that each

$$Y_i \sim \mathbf{N}(0, 1) \quad \text{independently.}$$

Then the hard-threshold estimator satisfies

$$\hat{\theta}_H = 0 \iff |Y_i| \leq \lambda \text{ for all } i = 1, \dots, n.$$

Equivalently,

$$P(\hat{\theta}_H = 0) = P(|Y_1| \leq \lambda, \dots, |Y_n| \leq \lambda).$$

Because the Y_i are independent, the joint probability factorises:

$$P(\hat{\theta}_H = 0) = \prod_{i=1}^n P(|Y_i| \leq \lambda) = [P(|X| \leq \lambda)]^n,$$

where $X \sim \mathbf{N}(0, 1)$.

Since $P(|X| \leq \lambda) = 1 - P(|X| > \lambda)$, we can rewrite this as

$$P(\hat{\theta}_H = 0) = (1 - P(|X| > \lambda))^n.$$

Let $\delta = P(|X| > \lambda)$. Clearly $0 < \delta < 1$ for any $\lambda > 0$. Then

$$1 - P(\hat{\theta}_H = 0) = 1 - (1 - \delta)^n.$$

We now apply *Bernoulli's inequality*, which states that for $n > 0$ and $0 < \delta < 1$,

$$1 - (1 - \delta)^n \leq n\delta.$$

Applying this to our expression gives

$$1 - P(\hat{\theta}_H = 0) \leq n\delta = nP(|X| > \lambda).$$

Using the bound from part (2.1),

$$P(|X| > \lambda) \leq \frac{2\varphi(\lambda)}{\lambda},$$

we obtain

$$1 - P(\hat{\theta}_H = 0) \leq n \frac{2\varphi(\lambda)}{\lambda}.$$

Now substitute $\lambda = \sqrt{2 \log n}$. Since $\varphi(\lambda) = (2\pi)^{-1/2} e^{-\lambda^2/2}$, we have

$$n \frac{2\varphi(\lambda)}{\lambda} = \frac{2n}{\lambda\sqrt{2\pi}} e^{-\lambda^2/2} = \frac{2n}{\sqrt{2\pi} 2 \log n} e^{-(2 \log n)/2} = \frac{2n}{\sqrt{4\pi \log n}} n^{-1} = \frac{1}{\sqrt{\pi \log n}}.$$

As $n \rightarrow \infty$, $\frac{1}{\sqrt{\pi \log n}} \rightarrow 0$. Hence

$$1 - P(\hat{\theta}_H = 0) \rightarrow 0 \quad \text{and therefore} \quad P(\hat{\theta}_H = 0) \rightarrow 1.$$

$$\boxed{P(\hat{\theta}_H = 0) \rightarrow 1 \quad \text{as } n \rightarrow \infty \text{ when } \lambda = \sqrt{2 \log n}.}$$

When the true mean vector θ is identically zero, each Y_i is a standard normal noise term. The threshold $\lambda = \sqrt{2 \log n}$ grows slowly with n so that the probability of even one coordinate exceeding λ vanishes asymptotically. Consequently, all coordinates are set to zero with probability tending to one.

Problem 3. Soft-Threshold Estimation

For $n = 1$, define the *soft-threshold* estimator

$$\hat{\theta}^S(Y) = \begin{cases} Y + \lambda, & Y < -\lambda, \\ 0, & |Y| \leq \lambda, \\ Y - \lambda, & Y > \lambda, \end{cases} \quad \lambda > 0,$$

where $Y \sim \mathcal{N}(\theta, 1)$. We study its risk

$$R_S(\theta) = \mathbb{E}_\theta[(\hat{\theta}^S - \theta)^2].$$

(3.1) Exact risk decomposition

The risk function can be decomposed according to the three regions of definition. Let's evaluate:

$$R_S(\theta) = \int_{-\lambda}^{\infty} (y - \lambda - \theta)^2 \varphi(y - \theta) dy + \int_{-\lambda}^{\lambda} \theta^2 \varphi(y - \theta) dy + \int_{-\infty}^{-\lambda} (y + \lambda - \theta)^2 \varphi(y - \theta) dy.$$

By making the change of variable

$$\varepsilon = y - \theta, \quad \text{so that } y = \theta + \varepsilon \quad \text{and} \quad dy = d\varepsilon.$$

Since $\varepsilon \sim \mathcal{N}(0, 1)$ under $Y \sim \mathcal{N}(\theta, 1)$, the integration limits shift by θ . The risk now becomes

$$R_S(\theta) = \int_{\lambda-\theta}^{\infty} (\varepsilon - \lambda)^2 \varphi(\varepsilon) d\varepsilon + \int_{-\lambda-\theta}^{\lambda-\theta} \theta^2 \varphi(\varepsilon) d\varepsilon + \int_{-\infty}^{-\lambda-\theta} (\varepsilon + \lambda)^2 \varphi(\varepsilon) d\varepsilon.$$

where $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ and $\varphi'(x) = -x\varphi(x)$.

We now compute each integral explicitly by integration by parts.

Bit 1.

$$I_1 := \int_{\lambda-\theta}^{\infty} (\varepsilon - \lambda)^2 \varphi(\varepsilon) d\varepsilon.$$

Expand the square and integrate term by term:

$$I_1 = \int_{\lambda-\theta}^{\infty} \varepsilon^2 \varphi(\varepsilon) d\varepsilon - 2\lambda \int_{\lambda-\theta}^{\infty} \varepsilon \varphi(\varepsilon) d\varepsilon + \lambda^2 \int_{\lambda-\theta}^{\infty} \varphi(\varepsilon) d\varepsilon.$$

Using the identities obtained by integration by parts ($\varphi'(x) = -x\varphi(x)$), gives:

$$\int_a^{\infty} \varepsilon \varphi(\varepsilon) d\varepsilon = \varphi(a), \quad \int_a^{\infty} \varepsilon^2 \varphi(\varepsilon) d\varepsilon = a \varphi(a) + 1 - \Phi(a), \quad \int_a^{\infty} \varphi(\varepsilon) d\varepsilon = 1 - \Phi(a),$$

with $a = \lambda - \theta$, we get

$$\begin{aligned} I_1 &= (\lambda - \theta) \varphi(\lambda - \theta) + 1 - \Phi(\lambda - \theta) - 2\lambda \varphi(\lambda - \theta) + \lambda^2 [1 - \Phi(\lambda - \theta)] \\ &= -(\theta + \lambda) \varphi(\lambda - \theta) + (1 + \lambda^2) [1 - \Phi(\lambda - \theta)]. \end{aligned}$$

Bit 2.

$$I_2 := \int_{-\infty}^{-\lambda-\theta} (\varepsilon + \lambda)^2 \varphi(\varepsilon) d\varepsilon.$$

Expanding and integrating each term:

$$I_2 = \int_{-\infty}^{-\lambda-\theta} \varepsilon^2 \varphi(\varepsilon) d\varepsilon + 2\lambda \int_{-\infty}^{-\lambda-\theta} \varepsilon \varphi(\varepsilon) d\varepsilon + \lambda^2 \int_{-\infty}^{-\lambda-\theta} \varphi(\varepsilon) d\varepsilon.$$

For integration over the left half-line, the identities are

$$\int_{-\infty}^b \varepsilon \varphi(\varepsilon) d\varepsilon = -\varphi(b), \quad \int_{-\infty}^b \varepsilon^2 \varphi(\varepsilon) d\varepsilon = \Phi(b) - b\varphi(b), \quad \int_{-\infty}^b \varphi(\varepsilon) d\varepsilon = \Phi(b).$$

With $b = -\lambda - \theta$, we find

$$\begin{aligned} I_2 &= [\Phi(b) - b\varphi(b)] + 2\lambda[-\varphi(b)] + \lambda^2\Phi(b) \\ &= (-b - 2\lambda)\varphi(b) + (1 + \lambda^2)\Phi(b). \end{aligned}$$

Since $b = -\lambda - \theta$, we have $-b - 2\lambda = \theta - \lambda$, and using $\varphi(b) = \varphi(\theta + \lambda)$, $\Phi(b) = \Phi(-\lambda - \theta)$, this becomes

$$I_2 = (\theta - \lambda)\varphi(\theta + \lambda) + (1 + \lambda^2)\Phi(-\lambda - \theta).$$

Bit 3.

$$I_3 := \theta^2 \int_{-\lambda-\theta}^{\lambda-\theta} \varphi(\varepsilon) d\varepsilon = \theta^2 [\Phi(\lambda - \theta) - \Phi(-\lambda - \theta)].$$

Summing the three parts. After combining the expressions for I_1 , I_2 , and I_3 , we obtain

$$\begin{aligned} R_S(\theta) &= -(\theta + \lambda)\varphi(\lambda - \theta) + (\theta - \lambda)\varphi(\theta + \lambda) \\ &\quad + (1 + \lambda^2)\{1 - \Phi(\lambda - \theta) + \Phi(-\lambda - \theta)\} + \theta^2[\Phi(\lambda - \theta) - \Phi(-\lambda - \theta)]. \end{aligned}$$

Exploiting the standard properties $\varphi(-x) = \varphi(x)$ and $\Phi(-x) = 1 - \Phi(x)$, the risk function takes the more compact form

$$\begin{aligned} R_S(\theta) &= -(\theta + \lambda)\varphi(\theta - \lambda) + (\theta - \lambda)\varphi(\theta + \lambda) \\ &\quad + (1 + \lambda^2)[\Phi(\theta - \lambda) + \Phi(-\lambda - \theta)] + \theta^2[\Phi(\lambda - \theta) - \Phi(-\lambda - \theta)]. \end{aligned}$$

We have thus established a closed-form expression for the soft-threshold estimator's risk in terms of the parameter θ and the threshold level λ .

(3.2) Bound at $\theta = 0$: showing that $R_S(0) \leq \frac{2\varphi(\lambda)}{\lambda}$

At $\theta = 0$, we have $Y \sim \mathcal{N}(0, 1)$ and, by symmetry of the estimator, the contributions from $y > \lambda$ and $y < -\lambda$ are equal. Therefore,

$$R_S(0) = 2 \int_{\lambda}^{\infty} (y - \lambda)^2 \varphi(y) dy.$$

Expanding the square gives

$$(y - \lambda)^2 = y^2 - 2\lambda y + \lambda^2,$$

hence

$$R_S(0) = 2 \left[\int_{\lambda}^{\infty} y^2 \varphi(y) dy - 2\lambda \int_{\lambda}^{\infty} y \varphi(y) dy + \lambda^2 \int_{\lambda}^{\infty} \varphi(y) dy \right].$$

To compute the three integrals, we use $\varphi'(y) = -y\varphi(y)$ and the definition $\Phi'(y) = \varphi(y)$.

1) $\int_{\lambda}^{\infty} \varphi(y) dy$. By definition of the cdf,

$$\int_{\lambda}^{\infty} \varphi(y) dy = 1 - \Phi(\lambda).$$

2) $\int_{\lambda}^{\infty} y \varphi(y) dy$. Since $-\varphi'(y) = y\varphi(y)$,

$$\int_{\lambda}^{\infty} y \varphi(y) dy = \int_{\lambda}^{\infty} (-\varphi'(y)) dy = [-\varphi(y)]_{\lambda}^{\infty} = \varphi(\lambda).$$

3) $\int_{\lambda}^{\infty} y^2 \varphi(y) dy$. Differentiate $y\varphi(y)$: $(y\varphi(y))' = \varphi(y) + y\varphi'(y) = \varphi(y) - y^2\varphi(y)$. Thus $y^2\varphi(y) = \varphi(y) - (y\varphi(y))'$, and

$$\int_{\lambda}^{\infty} y^2 \varphi(y) dy = \int_{\lambda}^{\infty} \varphi(y) dy - [y\varphi(y)]_{\lambda}^{\infty} = (1 - \Phi(\lambda)) - (0 - \lambda\varphi(\lambda)) = \lambda\varphi(\lambda) + 1 - \Phi(\lambda).$$

Using the Gaussian tail bound $1 - \Phi(\lambda) \leq \varphi(\lambda)/\lambda$ from Problem (2.1), we obtain

$$\begin{aligned} R_S(0) &\leq 2 \left[\lambda\varphi(\lambda) + 1 - \Phi(\lambda) - 2\lambda\varphi(\lambda) + \lambda^2 \frac{\varphi(\lambda)}{\lambda} \right] \\ &= 2(1 - \Phi(\lambda)) \leq \frac{2\varphi(\lambda)}{\lambda}. \end{aligned}$$

This establishes the required bound.

(3.3) Limiting risk as $\theta \rightarrow \infty$

From part (3.1) we have the identity

$$R_S(\theta) = -(\theta + \lambda) \varphi(\theta - \lambda) + (\theta - \lambda) \varphi(\theta + \lambda) + (1 + \lambda^2) [\Phi(\theta - \lambda) + \Phi(-\lambda - \theta)] + \theta^2 [\Phi(\lambda - \theta) - \Phi(-\lambda - \theta)]. \quad (\star)$$

We now take $\theta \rightarrow \infty$ term by term. Recall the standard limits

$$\Phi(x) \rightarrow 1 \ (x \rightarrow +\infty), \quad \Phi(x) \rightarrow 0 \ (x \rightarrow -\infty), \quad \varphi(x) \rightarrow 0 \ (|x| \rightarrow \infty).$$

Moreover, since $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ decays exponentially, *any* polynomial in θ grows more slowly; hence

$$(\theta \pm \lambda) \varphi(\theta \mp \lambda) \rightarrow 0, \quad \theta^2 \Phi(\lambda - \theta) \rightarrow 0, \quad \theta^2 \Phi(-\lambda - \theta) \rightarrow 0 \quad (\theta \rightarrow \infty).$$

Using these limits, substitute into $(\star)$:

$$\begin{aligned} \lim_{\theta \rightarrow \infty} R_S(\theta) &= \lim_{\theta \rightarrow \infty} \left\{ -(\theta + \lambda) \underbrace{\varphi(\theta - \lambda)}_{\rightarrow 0} + (\theta - \lambda) \underbrace{\varphi(\theta + \lambda)}_{\rightarrow 0} \right. \\ &\quad \left. + (1 + \lambda^2) \left[\underbrace{\Phi(\theta - \lambda)}_{\rightarrow 1} + \underbrace{\Phi(-\lambda - \theta)}_{\rightarrow 0} \right] + \theta^2 \left[\underbrace{\Phi(\lambda - \theta)}_{\rightarrow 0} - \underbrace{\Phi(-\lambda - \theta)}_{\rightarrow 0} \right] \right\} \\ &= \lim_{\theta \rightarrow \infty} \left\{ -(\theta + \lambda) \cdot 0 + (\theta - \lambda) \cdot 0 + (1 + \lambda^2)(1 + 0) + \theta^2(0 - 0) \right\} \\ &= 1 + \lambda^2. \end{aligned}$$

Therefore,

$$\lim_{\theta \rightarrow \infty} R_S(\theta) = 1 + \lambda^2.$$

(3.4) Comparison with the unbiased estimator $\hat{\theta} \equiv Y$

The unbiased estimator $\hat{\theta} = Y$ has constant risk

$$R_Y(\theta) = \mathbb{E}_\theta[(Y - \theta)^2] = \text{Var}(Y) = 1 \quad (\forall \theta),$$

so it serves as a benchmark: $R_Y(\theta) \equiv 1$.

For the soft-threshold estimator $\hat{\theta}_S$, from parts (3.2)–(3.3),

$$R_S(0) \leq \frac{2\varphi(\lambda)}{\lambda} \quad \text{and} \quad \lim_{|\theta| \rightarrow \infty} R_S(\theta) = 1 + \lambda^2.$$

Hence, since $R_S(0) < 1$ and $1 + \lambda^2 > 1$, and because $R_S(\theta)$ is continuous and even in θ , there exists a (unique) $c_\lambda > 0$ such that

$$R_S(c_\lambda) = 1, \quad R_S(\theta) < 1 \text{ iff } |\theta| < c_\lambda.$$

Thus, $\hat{\theta}_S$ dominates $\hat{\theta} = Y$ on $(-c_\lambda, c_\lambda)$ and is dominated outside it.

Using the exact expression

$$R_S(\theta) = -(\theta + \lambda)\varphi(\theta - \lambda) + (\theta - \lambda)\varphi(\theta + \lambda) + (1 + \lambda^2)[\Phi(\theta - \lambda) + \Phi(-\lambda - \theta)] + \theta^2[\Phi(\lambda - \theta) - \Phi(-\lambda - \theta)],$$

and solving $R_S(\theta) = 1$ for $\theta > 0$ gives:

λ	Region where $R_S(\theta) < 1$	Critical value c_λ
1	$ \theta \lesssim 1.2833$	$c_1 \approx 1.2833$
2	$ \theta \lesssim 1.0528$	$c_2 \approx 1.0528$
5	$ \theta \lesssim 1.0000$	$c_5 \approx 1.0000$

Hence, the soft-threshold estimator has smaller risk than the unbiased estimator for moderate signals ($|\theta| < c_\lambda$); for large signals $R_S(\theta) > 1$, and

$$\lim_{|\theta| \rightarrow \infty} R_S(\theta) = 1 + \lambda^2 > 1,$$

so $\hat{\theta} = Y$ is asymptotically more efficient.